

Manifold-Informed Architecture Benchmark

TINY_IMAGENET: 200 classes, 12,288D input

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waverider @ 4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07) | 2026-04-14 20:51:52
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Best: Intrinsic Dim (PCA→200D→output) test acc = 0.0336 ± 0.0012
vs Standard: +0.0070 (0.70 pp) 164.3× fewer params 207.8× acc/Kparam
Manifold: 12,288D → 20D (99.8% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	TINY_IMAGENET
Input dimensionality	12,288D
Classes	200
Intrinsic dim (d)	20
Variance threshold (τ)	0.9
Epochs	50
Trials	3

Manifold Discovery

Local PCA over the training set, $k = ?$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 200$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	19.6	0.8	17	21	99.8
0.90	16.6	0.9	14	18	99.9
0.85	14.2	1.0	11	16	99.9
0.80	12.3	1.0	9	14	99.9

Per-Class Intrinsic Dimensionality

Showing the 10 highest- and 10 lowest-dimensional classes of 200 total, sorted by mean d .

Class	Mean d	Std	Min	Max
146	20.0	0.0	20	20
55	19.9	0.3	19	20
13	19.5	0.5	19	20
99	19.5	0.5	19	20
108	19.5	0.5	19	20
173	19.2	0.4	19	20
90	19.1	0.3	19	20
147	19.1	0.7	18	20
152	19.1	0.3	19	20
16	19.0	0.4	18	20
...				
20	15.6	1.2	14	17
4	15.5	0.7	14	16
179	15.5	0.8	14	17
186	15.5	0.9	14	17
48	15.4	0.7	14	16
161	15.2	0.9	14	17
94	15.1	0.5	14	16
153	14.9	0.7	14	16
116	14.7	0.5	14	15
187	14.6	1.2	13	16

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Standard (1024→512)	13,211,336	0.0266 ± 0.0010	4565.6958	0.0000
Wide Manifold (d+1, d=200)	2,510,489	0.0241 ± 0.0019	64.0025	0.0000
Manifold (d=200)	2,498,000	0.0260 ± 0.0023	69.4988	0.0000
ManifoldAdam (1024→512, proj→200D)	13,211,336	0.0264 ± 0.0008	3297.0621	0.0000
PCA→200D + MLP (2d→d)	200,800	0.0279 ± 0.0025	228.4910	0.0001
Intrinsic Dim (PCA→200D→output)★	80,400	0.0336 ± 0.0012	6.8843	0.0004

★ = best performing architecture

Key Findings

- **Best architecture:** Intrinsic Dim (PCA→200D→output) — test accuracy 0.0336 ± 0.0012
- **vs Standard:** +0.0070 (0.70 pp) accuracy gain
- **Parameter reduction:** 164.3× fewer parameters (80,400 vs 13,211,336)
- **Parameter efficiency:** 207.8× improvement (0.0004 vs 0.0000 acc/Kparam)

- **Manifold compression:** 12,288D $\rightarrow$ 20D (99.8% of ambient dimensions carry no structure)

Result Figure

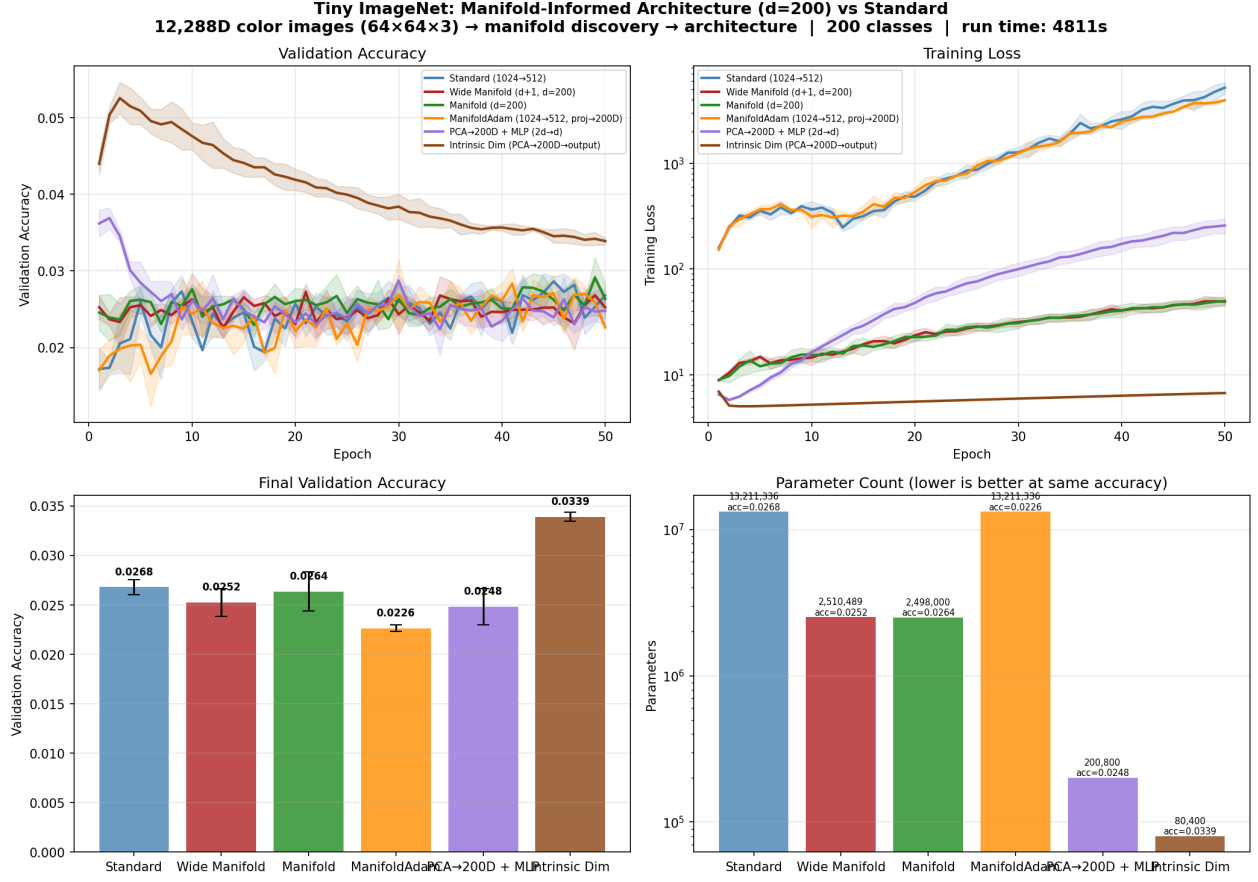


Figure 1: TINY_IMAGENET manifold-informed vs. standard architecture comparison. $d = 20, 50$ epochs, 3 trials.

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07)
Commit date	2026-04-14 20:51:52 -0400
Commit message	add: cifar10 results
Python	3.12.13
TensorFlow	2.16.2
Device	GPU
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
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